

Web Development Fundamentals

Assignment

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Section: AD(57)

1. Difference between Frontend, Backend and Full-stack Development.

Answer: **Frontend Development:**

Frontend development deals with the user interface of a website or application.

It includes HTML, CSS, and JavaScript.

Example:

Amazon website ka product page jo user dekhta hai, wo frontend ka part hai.

Backend Development:

Backend development handles server-side operations, databases, and application logic.

Example:

Amazon ka order processing system and payment handling backend ka part hai.

Full-stack Development:

Full-stack development includes both frontend and backend development.

Example:

A full-stack developer can create a complete e-commerce website including UI, server, and database.

2.Client-Server Model Diagram



3.Explain how a browser loads a web page from a server.

1. User enters a website URL in the browser.
2. Browser sends a request to the web server.
3. Server processes the request and sends HTML, CSS and JavaScript files.
4. Browser downloads these files.
5. Browser renders the webpage and displays it to the user.

4.List tools required for web development setup and their uses.

Tool	Purpose
VS Code	Code writing and editing
Browser	Testing websites
Git	Version control
GitHub	Code hosting
Node.js	Running JavaScript outside browser

5.Explain What a Web Server is with Examples.

A web server is software or hardware that stores, processes and delivers web pages to users.

Examples:

1. Apache
2. Nginx
3. Microsoft IIS

6.Explain the roles of frontend developer, backend developer, and database administrator.

Role	Work
Frontend Developer	Creates user interface
Backend Developer	Handles server and logic
Database Administrator	Manages database

7. Install VS Code and configure it for web development with a screenshot.

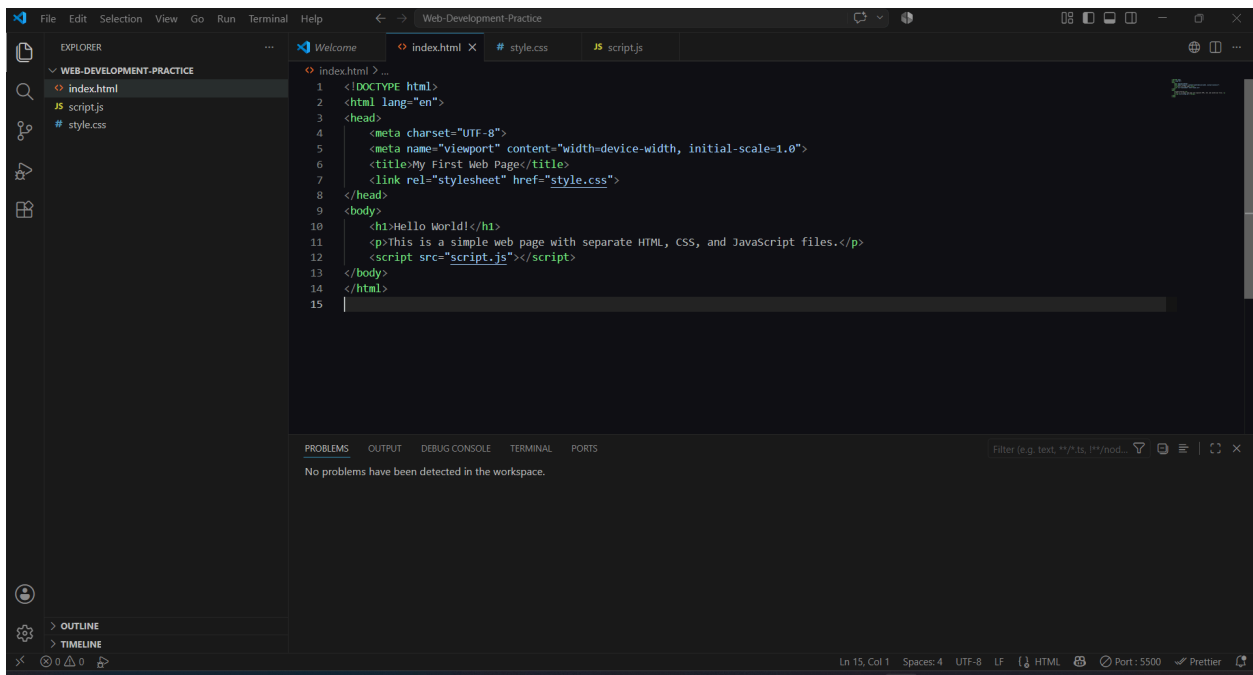
Visual Studio Code Setup:

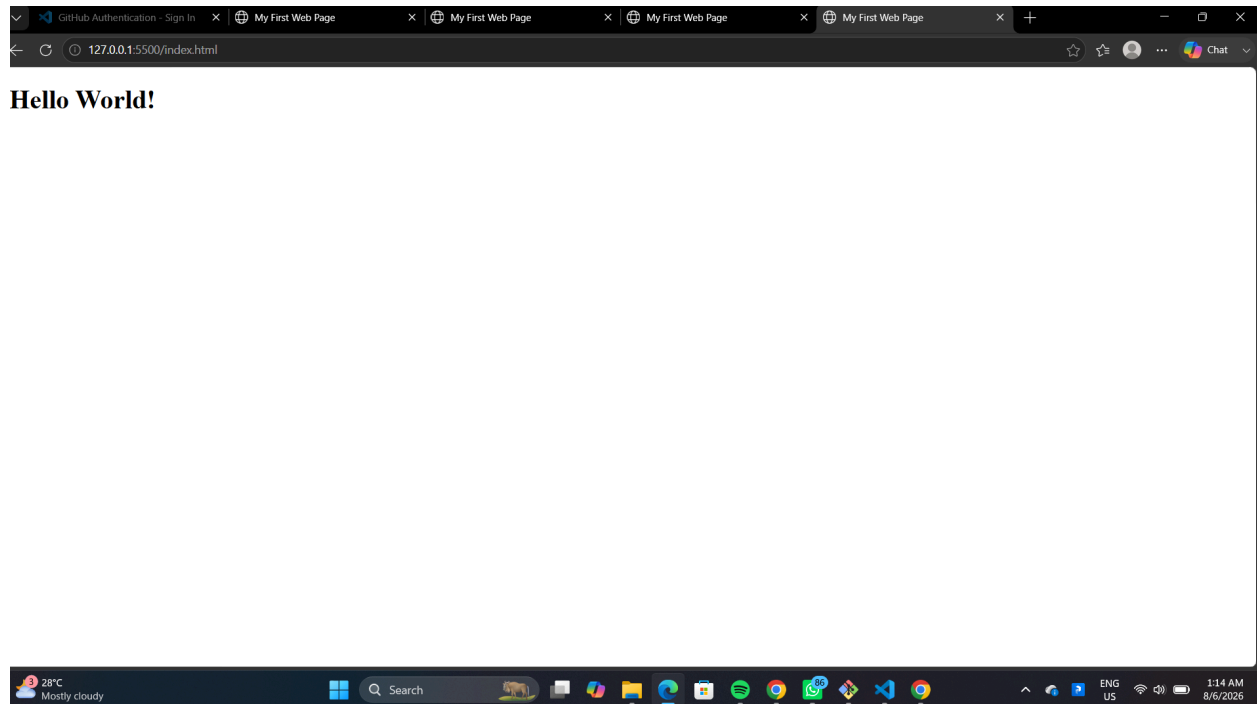
Visual Studio Code (VS Code) is a source code editor used for web development. It supports HTML, CSS, and JavaScript development with the help of extensions.

Steps performed:

1. Installed Visual Studio Code for web development.
2. Created an HTML file named [index.html](#).
3. Created CSS and JavaScript files for styling and functionality.
4. Installed useful extensions:
 - [Live Server](#)
 - [Prettier - Code Formatter](#)
5. Opened the HTML file using Live Server.
6. Successfully displayed the "My First Webpage" in the browser.

Screenshot of VS Code Setup:





8.Explain static and dynamic websites with examples.

Static Website

Fixed content

HTML/CSS based

Example: Portfolio website

Dynamic Website

Content changes

Uses backend and database

Example: Instagram

9.List five web browsers and their rendering engines.

Browser

Rendering Engine

Chrome

Blink

Firefox

Gecko

Safari

WebKit

Edge

Blink

Opera

Blink

10. Draw a basic web architecture diagram with client, server, database, and APIs.

